Weather and Air Quality in Bengaluru, 2023-2025: A Data Visualization Study

A PROJECT REPORT

Submitted by

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for the course

CS2214 Introduction to Data Analysis

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DECLARATION

We, the students listed below, of the 3rd semester **B.Tech in Computer Science & Engineering** at the School of Computer Science and Engineering, **RV University**, hereby declare that the project work titled "**Weather and Air Quality in Bengaluru, 2023-2025: A Data Visualization Study**" has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science & Engineering** during the academic year **2026-2027**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

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CERTIFICATE

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TABLE OF CONTENTS

	TITLE	
	LIST OF FIGURES	**i**
	LIST OF SYMBOLS AND ABBREVIATIONS	**ii**
1.0	**INTRODUCTION**	**1**
1.1	Background	1
1.2	Problem Statement	1
1.3	Objectives	1
1.4	Scope and Limitations	1
1.5	Organisation of the Report	2
2.0	**DATASET AND METHODOLOGY**	**3**
2.1	Data Source and Selection	3
2.2	Location, Period and Sampling Approach	3
2.3	Dataset Description	3
2.4	Data Preparation and Cleaning	4
2.5	Derived Variables and Benchmarks	4
2.6	Tools and Reproducibility	5
3.0	**METHOD OF ANALYSIS**	**6**
3.1	Descriptive Statistics	6
3.2	Seasonal and Monthly Aggregation	6
3.3	Comparison of Rainy and Dry Days	6
3.4	Correlation Analysis	6
3.5	Comparison with Benchmarks	6
3.6	Visualization Design	6
4.0	**RESULTS AND DISCUSSION**	**8**
4.1	Overview of the Data	8
4.2	Seasonal Pattern of Weather	9
4.3	Air Quality Through the Years	10
4.4	PM2.5 by Month and Rainfall	11
4.5	Rainy Days and Dry Days	12
4.6	Air-Quality Categories by Season	13
4.7	Relationships Between Variables	14
4.8	Main Findings	14
4.9	Recommendations	15
4.10	Limitations of the Study	15
5.0	**CONCLUSION**	**16**
6.0	**FUTURE SCOPE**	**17**
	REFERENCES	**18**
	APPENDIX A: MONTHLY SUMMARY	**19**
	APPENDIX B: PROJECT FILES AND SOURCE CODE	**19**

LIST OF FIGURES

Figure 4.1	Monthly temperature, humidity and rainfall	9
Figure 4.2	Calendar heatmap of daily AQI	10
Figure 4.3	Daily PM2.5 and 30-day average	10
Figure 4.4	Mean PM2.5 by month and average yearly rainfall	11
Figure 4.5	PM2.5 on dry and rainy days	12
Figure 4.6	AQI categories by season	13
Figure 4.7	Spearman correlation heatmap	14

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / abbreviation	Meaning
AQI	Air Quality Index (here, the US AQI, scale 0-500)
API	Application Programming Interface
CAMS	Copernicus Atmosphere Monitoring Service
CPCB	Central Pollution Control Board, India
CSV	Comma-separated values (file format)
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA5	Fifth-generation ECMWF atmospheric reanalysis
IMD	India Meteorological Department
IQR	Interquartile range
NAAQS	National Ambient Air Quality Standards (India)
NO_2	Nitrogen dioxide
O_3	Ozone
PM2.5	Particulate matter with diameter below 2.5 µm
PM10	Particulate matter with diameter below 10 µm
ρ (rho)	Spearman rank correlation coefficient
SD	Standard deviation
USN	University Seat Number
US EPA	United States Environmental Protection Agency
WHO	World Health Organization
µg/m³	Micrograms per cubic metre (unit of pollutant concentration)
°C	Degrees Celsius
mm	Millimetres (rainfall depth)

1.0 INTRODUCTION

1.1 Background

Temperature, humidity and rainfall describe the weather. Particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2) and ozone (O_3) describe the air people breathe. The two are connected in several ways: rain can remove particles from the air, and heat and sunlight help ozone form. Looking at both together, day by day, shows when the air is cleaner or dirtier and which weather goes with each.

Bengaluru sits about 914 m above sea level and has a moderate climate with a clear monsoon season. Two benchmarks help judge the air. The World Health Organization (WHO) 2021 guideline for the 24-hour mean of PM2.5 is 15 µg/m³, and India's National Ambient Air Quality Standards (NAAQS) set a 24-hour limit of 60 µg/m³. We compare the observed levels with both.

1.2 Problem Statement

Using public daily data for Bengaluru from January 2023 to December 2025, we ask three questions. How do temperature, humidity and rainfall change through the year? How does air quality change alongside them? And in which seasons is the air worst compared with health guidelines?

1.3 Objectives

- Describe the seasonal pattern of temperature, humidity and rainfall in Bengaluru.
- Measure how PM2.5, PM10, NO_2, ozone and the US AQI vary by month and by season.
- Compare air quality on rainy days and dry days.
- Check the observed air quality against WHO and CPCB benchmarks.
- Present the results as labelled Python charts, with evidence-based findings and practical recommendations.

1.4 Scope and Limitations

The study covers one location (Bengaluru), three years (1,096 days) and ten daily variables. The data are modelled grid values from a public service. They are not readings from a ground monitor, and our group did not collect them. The analysis is observational, so it shows patterns and associations but cannot show cause and effect. Three years are enough to describe the yearly cycle and too few to judge long-term trends.

1.5 Organisation of the Report

Chapter 2 covers the dataset and how we prepared it, and Chapter 3 the analysis methods. Chapter 4 gives the results, findings and recommendations. Chapters 5 and 6 are the conclusion and the future scope.

2.0 DATASET AND METHODOLOGY

2.1 Data Source and Selection

We did not collect these data. We downloaded them from two public application programming interfaces (APIs) run by Open-Meteo (open-meteo.com). Open-Meteo states that its data are licensed under CC BY 4.0, so we credit it here and on the poster. Table 2.1 lists the two products.

Table 2.1: Data sources used

Product	Underlying data	Variables used	Resolution
Open-Meteo Historical Weather API	ERA5 reanalysis (ECMWF)	Temperature at 2 m, relative humidity at 2 m, precipitation	Hourly
Open-Meteo Air Quality API	CAMS atmospheric composition model (Copernicus / ECMWF)	PM2.5, PM10, NO_2, ozone, US AQI	Hourly

We chose Open-Meteo because the topic asks for temperature, humidity, rainfall and air-quality readings from reliable sources, and Open-Meteo has all four for the same place and dates. The format is consistent, no access key is needed, and anyone can repeat the download. Its inputs, ERA5 and CAMS, come from established meteorological centres.

2.2 Location, Period and Sampling Approach

- Location: Bengaluru, India (latitude 12.97° N, longitude 77.59° E, elevation about 914 m).
- Period: 1 January 2023 to 31 December 2025, which is 1,096 days.
- Time zone: Asia/Kolkata (IST).
- Download date: 11 November 2026.
- Sampling approach: none. We downloaded every hourly value for the period and combined them into one row per day.

2.3 Dataset Description

The prepared dataset has 1,096 daily records and 10 attributes, with no missing values. Each attribute uses a single unit throughout. Table 2.2 gives the meaning, unit and daily aggregation of each one.

Table 2.2: Attributes of the daily dataset

Attribute	Meaning	Unit	Daily value from hourly data
date	Calendar date (YYYY-MM-DD)	-	-
temp_mean	Air temperature at 2 m	°C	mean
temp_max	Highest hourly temperature	°C	maximum
temp_min	Lowest hourly temperature	°C	minimum
humidity_mean	Relative humidity at 2 m	%	mean
rainfall_mm	Precipitation	mm	sum
pm25	Particulate matter below 2.5 µm	µg/m³	mean
pm10	Particulate matter below 10 µm	µg/m³	mean
no2	Nitrogen dioxide	µg/m³	mean
ozone	Ozone	µg/m³	mean
aqi_us	US Air Quality Index	0-500	mean of hourly values

2.4 Data Preparation and Cleaning

We joined the two hourly series on the date and aggregated them to daily values as shown in Table 2.2. We then ran the checks in Table 2.3, which also shows how each was handled.

Table 2.3: Data-quality checks and handling

Check	Result	Handling
Missing values	0 of 10,960 cells	None needed
Duplicate dates	0	None needed
Temperature order (min ≤ mean ≤ max)	Holds on all days	None needed
Humidity between 0 and 100 %	All values valid	None needed
Rainfall and pollutants not negative	All values valid	None needed
Temperature between −10 and 50 °C	All values valid	None needed
Outliers (1.5 × IQR rule)	Present: 140 days for rainfall, 42 for mean temperature, 48 for NO_2, 2 for PM2.5 and 4 for PM10	Kept. They are real extremes, not errors
Units	One unit per column	None needed

Rainfall has many outliers under the 1.5 × IQR rule because most days have little or no rain, so any wet day sits far above the interquartile range. The highest daily maximum temperature was 39.5 °C on 1 May 2024. We kept all of these values. The dataset has no personal or sensitive information.

2.5 Derived Variables and Benchmarks

We derived four variables.

- Season: Winter (December to February), Summer (March to May), Monsoon (June to September) and Post-monsoon (October to November).

- AQI category: the US AQI band of each day. Good (0-50), Moderate (51-100), Unhealthy for sensitive groups (101-150) and Unhealthy (151-200). No day reached 201 or above.
- Rainy day: a day with 2.5 mm of rain or more. All other days are dry days.
- Year and month: taken from the date.

Table 2.4: Air-quality benchmarks used

Pollutant	WHO 2021 guideline (24-hour)	CPCB NAAQS limit (24-hour)
PM2.5	15 µg/m³	60 µg/m³
PM10	45 µg/m³	100 µg/m³

2.6 Tools and Reproducibility

We used Python 3 throughout: pandas and NumPy for data handling, SciPy for the statistical test and Matplotlib for every chart. The whole workflow, from download to charts, is in a Google Colab notebook (*Bengaluru_Weather_AQ_Visualization.ipynb*) and in the project's scripts, so every number in this report can be regenerated. Appendix B lists the files.

3.0 METHOD OF ANALYSIS

3.1 Descriptive Statistics

For each variable we computed the mean, standard deviation, minimum, median and maximum over the 1,096 days (Table 4.1). These show the typical level and spread before any comparison.

3.2 Seasonal and Monthly Aggregation

We grouped the daily values by season and by calendar month. In the month-wise charts, rainfall is the average yearly total for that month (the three-year total divided by three) and the other variables are three-year means. Grouping smooths day-to-day noise and shows the yearly cycle.

3.3 Comparison of Rainy and Dry Days

We compared daily PM2.5 on rainy days (2.5 mm or more) and dry days using medians and a box plot. A two-sided Mann-Whitney U test gave an indicative check of whether the two groups differ; it does not assume normally distributed data. Consecutive days are not fully independent, so we read the p-value as a guide to the size of the difference and not as an exact probability.

3.4 Correlation Analysis

We measured relationships between variables with the Spearman rank correlation coefficient (ρ). We chose it over Pearson's coefficient because rainfall is heavily skewed with many zeros, and because the relationships only need to be monotonic, not linear. A value near +1 means two variables tend to rise together, near −1 means one tends to fall as the other rises, and near 0 means no consistent link. Correlation describes association only.

3.5 Comparison with Benchmarks

We counted the days above the WHO and CPCB limits (Table 2.4) for PM2.5 and PM10, and the days with a US AQI above 100, the point where air quality is rated unhealthy for sensitive groups.

3.6 Visualization Design

We drew seven charts with Matplotlib. Each has a title, labelled axes with units and readable scales, and a legend wherever more than one colour or line appears. The colours stay distinguishable for readers with colour-vision deficiency: one blue ramp for magnitude, blue

and orange for two groups, and a blue-red scale centred on grey for correlations. No chart uses two vertical axes; variables with different units get separate panels. Table 3.1 lists the chart types and their purposes.

Table 3.1: Charts used

Figure	Chart type	Purpose
4.1	Line plot with band, and bar chart	Monthly temperature, humidity and rainfall
4.2	Calendar heatmap	Daily AQI across three years
4.3	Line plot	Daily PM2.5 and its 30-day average
4.4	Bar chart (two panels)	Monthly PM2.5 against the WHO guideline, with rainfall
4.5	Box plot	PM2.5 distribution on dry and rainy days
4.6	Stacked bar chart	Share of days in each AQI category, by season
4.7	Correlation heatmap	Relationships between weather and pollutants

4.0 RESULTS AND DISCUSSION

4.1 Overview of the Data

Table 4.1: Descriptive statistics of the daily data (1,096 days)

Variable	Unit	Mean	SD	Min	Median	Max
Mean temperature	°C	23.6	2.2	18.4	23.2	31.5
Maximum temperature	°C	29.3	3.0	21.4	28.6	39.5
Minimum temperature	°C	19.1	2.0	11.9	19.6	24.7
Relative humidity	%	68.8	15.2	27.2	73.3	94.0
Rainfall (daily total)	mm	2.7	5.0	0.0	0.3	36.6
PM2.5	µg/m³	20.9	12.7	2.0	20.2	66.3
PM10	µg/m³	28.9	16.7	2.8	28.0	107.1
NO_2	µg/m³	16.3	8.0	5.1	14.1	51.0
Ozone	µg/m³	70.5	25.7	19.0	72.3	143.4
US AQI	index	71.8	29.7	16.8	72.7	155.6

Daily mean temperature averaged 23.6 °C, with a hottest daily maximum of 39.5 °C (1 May 2024) and a coldest daily minimum of 11.9 °C (11 January 2023). Humidity ranged widely, from 27 % to 94 %. The heaviest rain in one day was 36.6 mm (30 May 2023). Mean PM2.5 was 20.9 µg/m³, with a highest daily value of 66.3 µg/m³ on 26 October 2024. The highest daily AQI was 156 on 2 May 2025. Yearly rainfall totals were 796 mm (2023), 1081 mm (2024) and 1035 mm (2025).

4.2 Seasonal Pattern of Weather

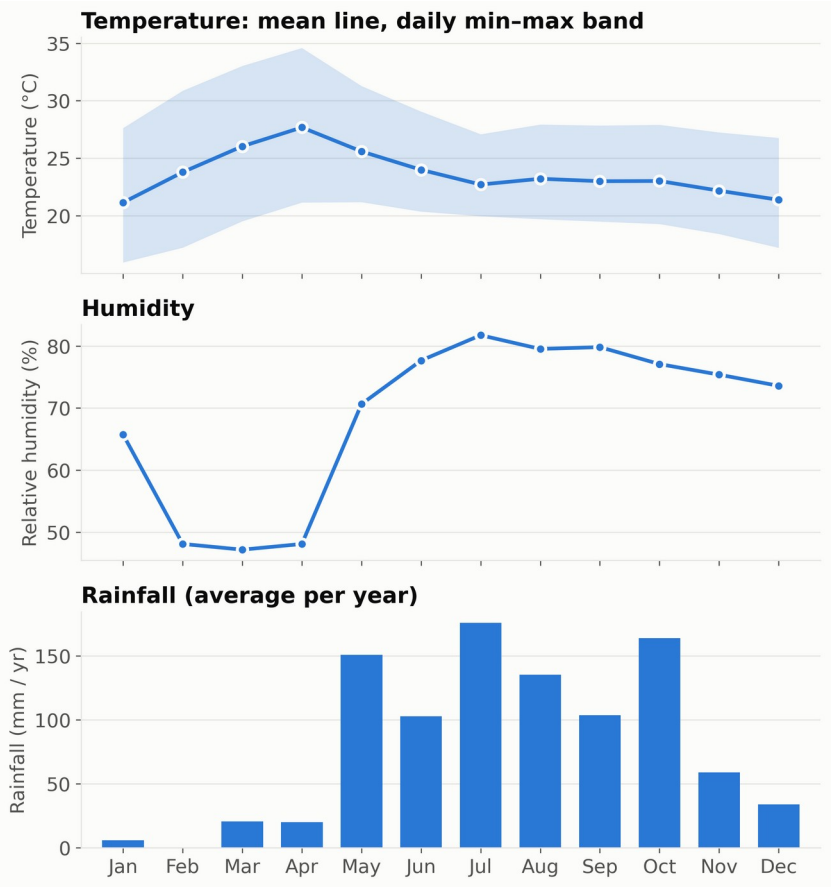


Figure 4.1: Monthly temperature, humidity and rainfall in Bengaluru, 2023-2025 (temperature in °C, humidity in %, rainfall in mm per year).

Figure 4.1 shows the yearly cycle. Mean temperature peaks in April (27.7 °C) and is lowest in January (21.1 °C). The gap between the daily minimum and maximum is widest in the dry months February to April, when the daily maximum averages up to 34.6 °C. Relative humidity is lowest in March (47 %) and highest in July (82 %). Rain falls mostly from May to October, with almost none in February. The Monsoon brings 53 % of the yearly rain (on average 517 mm), against 223 mm in Post-monsoon, 191 mm in Summer and 39 mm in Winter (Table 4.2).

Table 4.2: Seasonal means (rainfall is the average yearly total for the season)

Season	Temp (°C)	Humidity (%)	Rain (mm)	PM2.5	PM10	NO_2	Ozone	AQI
Winter	22.0	63	39	28.5	36.4	16.2	87	88
Summer	26.4	55	191	26.6	37.4	19.3	92	89
Monsoon	23.2	80	517	9.0	14.9	11.9	44	43
Post-monsoon	22.6	76	223	24.9	33.2	20.8	66	80

PM2.5, PM10, NO_2 and ozone are in µg/m³.

4.3 Air Quality Through the Years

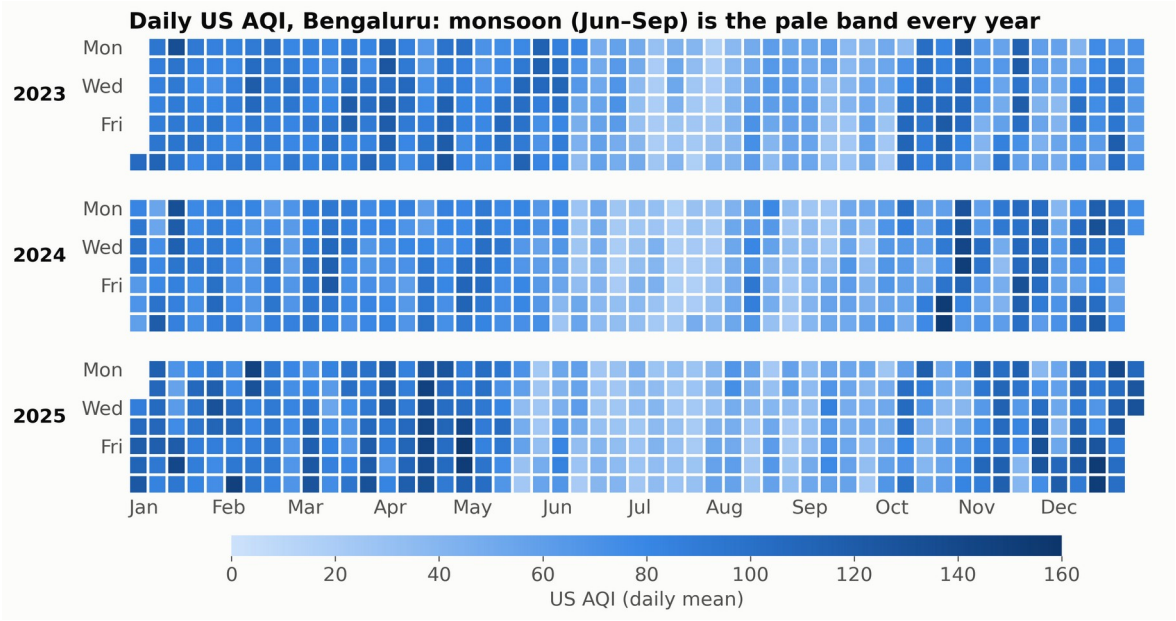


Figure 4.2: Calendar heatmap of the daily US AQI in Bengaluru, 2023-2025. Each square is one day; paler squares mean lower AQI.

In Figure 4.2 the paler squares of the monsoon months, June to September, form a band that repeats every year, while darker squares dominate January to May and October to December. The highest AQI day was 2 May 2025 (AQI 156). Across the whole period the AQI reached the "unhealthy" band (151-200) on only 7 days.

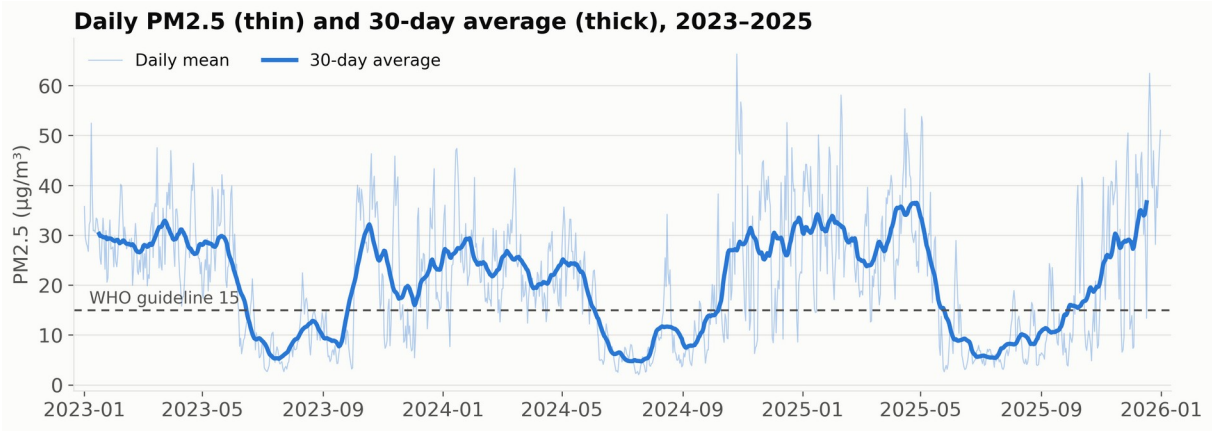


Figure 4.3: Daily PM2.5 (thin line) and its 30-day centred average (thick line), with the WHO guideline of 15 µg/m³ (dashed).

Figure 4.3 shows the same cycle for PM2.5. The 30-day average falls below the WHO guideline in each monsoon and rises above it for the rest of the year. All 86 days above 40 µg/m³ fall outside the monsoon. The highest day, 26 October 2024, reached 66.3 µg/m³. Yearly mean PM2.5 was 21.4, 19.5 and 21.8 µg/m³ in 2023, 2024 and 2025. These are close, so the three years show a stable seasonal pattern but no clear long-term trend.

4.4 PM2.5 by Month and Rainfall

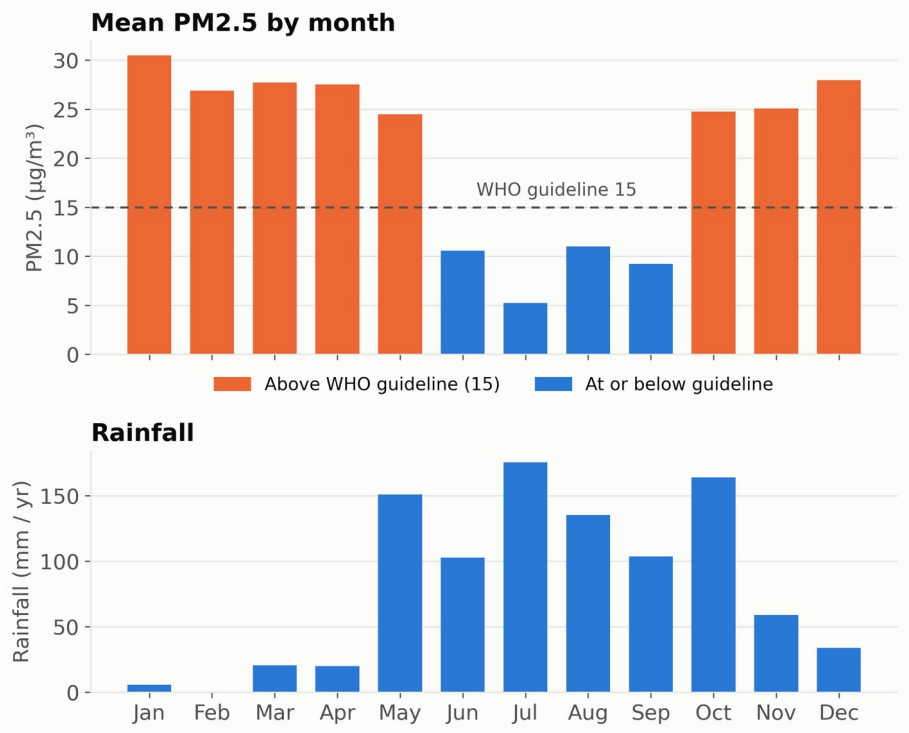


Figure 4.4: Mean PM2.5 by calendar month against the WHO guideline (top) and average yearly rainfall by month (bottom).

Only June to September have a mean PM2.5 at or below 15 µg/m³ (Figure 4.4). July is lowest at 5.2 µg/m³ and January highest at 30.5 µg/m³. The monthly mean is above the WHO guideline in 8 of the 12 months. May stands out: it has heavy rain (about 151 mm per year), yet its PM2.5 (24.5 µg/m³) stays above the guideline. Rainfall alone therefore does not explain the monthly pattern. Wind and the mixing of air, which also matter, are not in this dataset.

4.5 Rainy Days and Dry Days

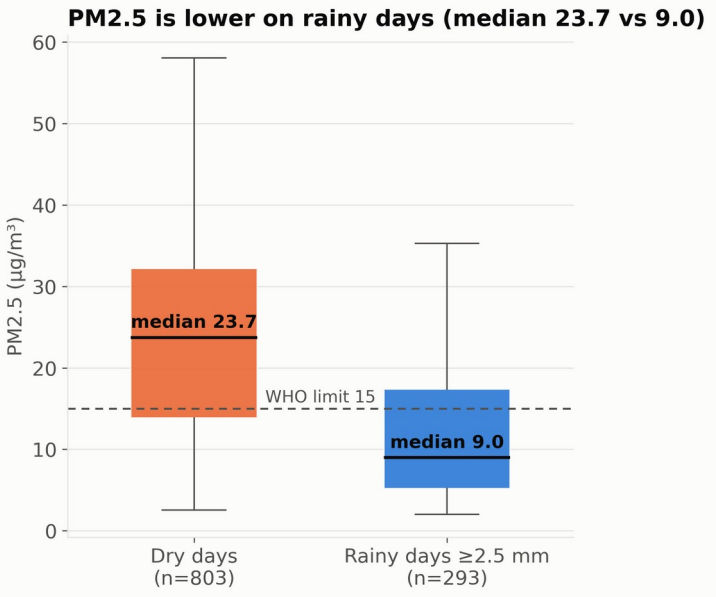


Figure 4.5: Distribution of daily PM2.5 on dry days and rainy days (rainy: 2.5 mm or more). Boxes show the interquartile range; the line is the median.

Table 4.3: Air quality on dry and rainy days

Group	Days	PM2.5 median	PM2.5 mean	PM10 mean	AQI mean	Humidity (%)
Dry days	803	23.7	23.8	32.9	79	64
Rainy days	293	9.0	12.9	18.1	53	82

PM2.5 and PM10 in µg/m³.

The median PM2.5 was 9.0 µg/m³ on rainy days and 23.7 µg/m³ on dry days, so dry days ran 2.6 times higher at the median (Figure 4.5, Table 4.3). Most rainy days lie below the WHO guideline, and dry days spread much wider and reach higher values. A Mann-Whitney U test also finds a clear difference between the two groups ($p < 0.001$). The Spearman correlation between daily rainfall and PM2.5 is −0.58, and the correlation with the previous day's rainfall is −0.57. Rain goes with cleaner air here, but this is an association: rainy days also differ in humidity, cloud and wind in ways this dataset does not record.

4.6 Air-Quality Categories by Season

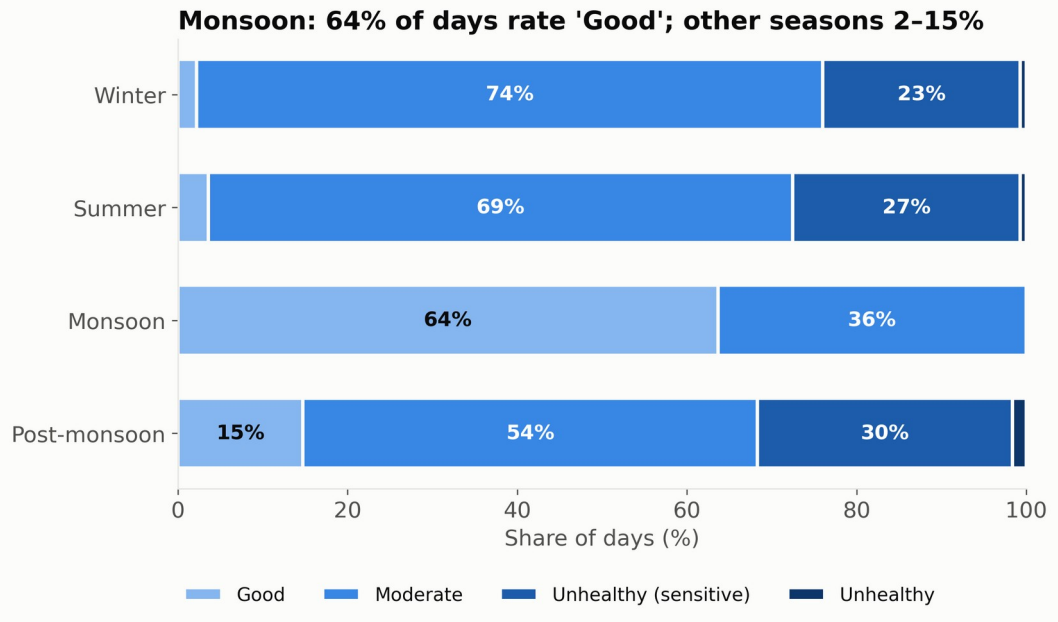


Figure 4.6: Share of days in each US AQI category, by season.

In the Monsoon, 64 % of days (233 of 366) were rated Good and none rose above Moderate (Figure 4.6). In the other three seasons only 6 % of days (43 of 730) were rated Good. Moderate days made up 54 % to 74 % of those seasons, and 23 % to 30 % of days were rated Unhealthy for sensitive groups. Table 4.4 gives the day counts and Table 4.5 compares the days with the benchmarks.

Table 4.4: Days in each AQI category, by season

Season	Good	Moderate	Unhealthy (sensitive)	Unhealthy	Total
Winter	6	200	63	2	271
Summer	10	190	74	2	276
Monsoon	233	133	0	0	366
Post-monsoon	27	98	55	3	183

Table 4.5: Days above air-quality benchmarks (1,096 days)

Measure	Days	Share of days
PM2.5 above WHO guideline (15 µg/m³)	675	62 %
PM2.5 above CPCB limit (60 µg/m³)	2	0.2 %
PM10 above WHO guideline (45 µg/m³)	197	18 %
PM10 above CPCB limit (100 µg/m³)	1	0.1 %
US AQI above 100	199	18 %

PM2.5 passed the Indian limit on only 2 days, but it passed the stricter WHO guideline on 62 % of days. All 199 days with an AQI above 100 fell outside the monsoon.

4.7 Relationships Between Variables

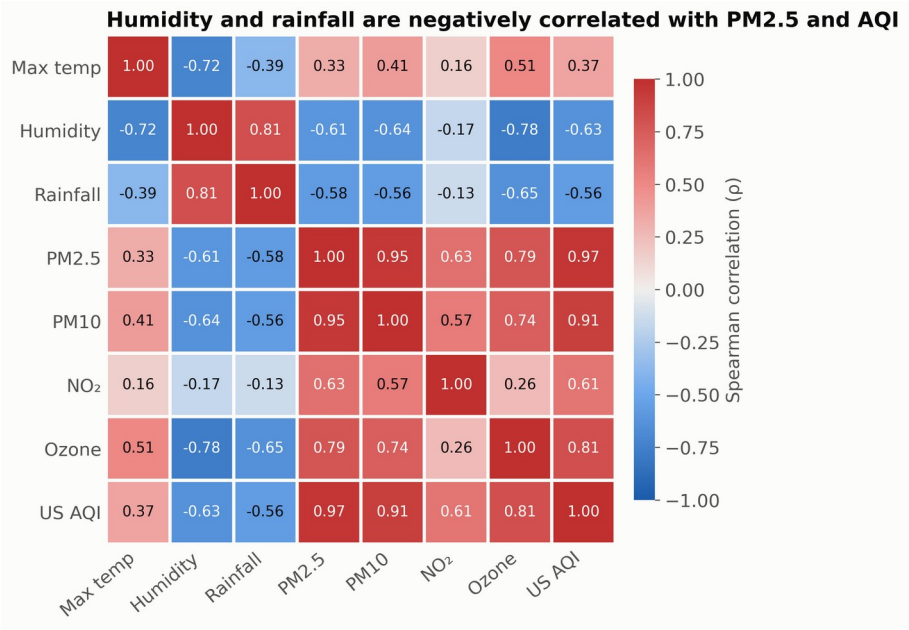


Figure 4.7: Spearman rank correlation between weather variables and pollutants (daily data, 1,096 days).

Table 4.6: Selected Spearman correlations (ρ)

Pair of variables	ρ
Rainfall and PM2.5	−0.58
Humidity and PM2.5	−0.61
Humidity and ozone	−0.78
Maximum temperature and ozone	0.51
Maximum temperature and humidity	−0.72
PM2.5 and PM10	0.95
PM2.5 and NO_2	0.63

Humidity and rainfall run opposite to the particles (Figure 4.7). Humidity correlates with PM2.5 at −0.61 and with ozone at −0.78. Ozone rises with the daily maximum temperature (ρ = 0.51), and mean ozone is 92 µg/m³ in Summer against 44 µg/m³ in the Monsoon. PM2.5 and PM10 move almost together (0.95), and PM2.5 closely tracks the AQI (0.97), because particles most often set the AQI. NO_2 is only weakly related to humidity and rainfall (−0.17 and −0.13). All of these are correlations in observational data, so they do not show that one variable causes another.

4.8 Main Findings

- Finding 1. The monsoon air was cleanest. 64 % of June to September days were rated Good against 6 % of other days. Mean PM2.5 was 9 µg/m³ in the Monsoon and 27 µg/m³ in the rest of the year.

- Finding 2. PM2.5 was lower on rainy days. The median was 9.0 µg/m³ on rainy days against 23.7 µg/m³ on dry days (Spearman $\rho = -0.58$ between rainfall and PM2.5; $p < 0.001$).
- Finding 3. PM2.5 exceeded the WHO guideline on most days. It was above 15 µg/m³ on 675 of 1,096 days (62 %) and in 8 of 12 months, but above the Indian limit of 60 µg/m³ on only 2 days.
- Finding 4. Days with a poor AQI fall in the dry months. All 199 days (18 %) with an AQI above 100 came between October and May.
- Finding 5. The yearly cycle is stable. Mean PM2.5 was 21.4, 19.5 and 21.8 µg/m³ in the three years, with the same seasonal shape each year.

4.9 Recommendations

- Recommendation 1. Time health advisories to the dry months. All 199 days with an AQI above 100 came between October and May, so advisories for sensitive groups (children, older adults and people with asthma) and the planning of outdoor events should focus on those months.
- Recommendation 2. Do not rely on rain to clean the air. The air meets the WHO guideline only in the monsoon months, so plans to cut emissions and dust should cover October to May. Data on likely sources (traffic, construction, burning) should be collected to decide where to act.
- Recommendation 3. Check the data against ground stations before using it for policy. The values here are modelled, so a comparison with CPCB monitor readings over five or more years should confirm the patterns before they guide decisions.

4.10 Limitations of the Study

- The data are modelled grid values for one point. They can differ from a single ground monitor and miss very local conditions.
- Only three years are covered, which is too short to separate seasonal variation from long-term trends.
- Correlations describe association only. Wind, traffic, industry and burning also affect air quality and are not in the dataset.
- Daily averages hide hourly peaks, such as rush-hour pollution, that matter for health.

5.0 CONCLUSION

We analysed 1,096 days of public weather and air-quality data for Bengaluru (2023-2025). The weather followed a clear yearly cycle: temperature peaked in April, humidity was lowest in the dry months, and rain fell mostly from May to October. Air quality followed the seasons too. It was cleanest in the monsoon (June to September) and above the WHO PM2.5 guideline in the other eight months. PM2.5 was clearly lower on rainy days (median 9.0 against 23.7 $\mu g/m^3$), and all 199 days with a poor AQI fell outside the monsoon.

We met all five objectives. We described the seasonal weather pattern, measured air quality by month and season, compared rainy and dry days, checked the values against WHO and CPCB benchmarks, and drew seven labelled charts. The findings are associations in observational, modelled data, and they do not show that rain makes the air cleaner. They do support the recommendations to time health advisories and plan emission control around the dry months.

6.0 FUTURE SCOPE

- Validate with ground stations. Compare the modelled values with CPCB monitor readings in Bengaluru to measure the difference.
- Extend the period and the places. Five to ten years of data from several cities would separate trends from the seasonal cycle.
- Add explanatory variables. Wind speed, boundary-layer height, traffic counts and fire activity could explain why May is rainy yet above the guideline.
- Use stronger models. Time-series or regression models can allow for the dependence between consecutive days, and can test whether rainfall adds explanatory power once season is controlled for.
- Look at hourly data. Hourly values would show daily cycles and rush-hour peaks that daily averages hide.
- Build an interactive dashboard. It could show the current day against the WHO and CPCB limits and the seasonal pattern.

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APPENDIX A: MONTHLY SUMMARY

Table A.1: Monthly means over 2023-2025 (rainfall is the average yearly total for the month)

Month	Temp mean (°C)	Temp max (°C)	Temp min (°C)	Humidity (%)	Rain (mm)	PM2.5 (µg/m³)	AQI
January	21.1	27.6	15.9	66	6	30.5	92
February	23.8	30.9	17.2	48	0	26.9	87
March	26.0	33.0	19.5	47	20	27.7	90
April	27.7	34.6	21.1	48	20	27.5	93
May	25.6	31.3	21.2	71	151	24.5	83
June	24.0	29.1	20.4	78	103	10.6	48
July	22.7	27.1	20.0	82	176	5.2	30
August	23.2	27.9	19.7	80	135	11.0	49
September	23.0	27.8	19.5	80	103	9.2	45
October	23.0	27.9	19.3	77	164	24.8	79
November	22.2	27.2	18.4	75	59	25.1	81
December	21.4	26.8	17.2	74	34	27.9	86

APPENDIX B: PROJECT FILES AND SOURCE CODE

Table B.1: Files submitted with this report

File	Content
data/bengaluru_daily_2023_2025.csv	Daily dataset (10 columns, 1,096 rows)
data/bengaluru_daily_clean.csv	Dataset with derived columns (season, AQI category, rainy-day flag)
data/raw/weather.json, aq.json	Hourly data exactly as returned by the APIs
data/DATA_DICTIONARY.md	Source, period, attribute meanings, quality checks
Bengaluru_Weather_AQ_Visualization.ipynb	Google Colab notebook: download, preparation, analysis and all charts
scripts/build_daily.py, analyse.py, make_figures.py	Python scripts for aggregation, analysis and the figures in this report
poster/	Poster (PDF) and its source files

Core of the data-preparation step (hourly values to daily values):

```
daily_w = weather.groupby("date").agg(
    temp_mean=("temperature_2m", "mean"), temp_max=("temperature_2m", "max"),
    temp_min=("temperature_2m", "min"), humidity_mean=("relative_humidity_2m",
"mean"),
    rainfall_mm=("precipitation", "sum"))
daily_a = air.groupby("date").agg(
    pm25=("pm2_5", "mean"), pm10=("pm10", "mean"), no2=("nitrogen_dioxide",
"mean"),
    ozone=("ozone", "mean"), aqi_us=("us_aqi", "mean"))
df = daily_w.join(daily_a, how="outer").round(2)
assert df.isna().sum().sum() == 0
df["rainy"] = df.rainfall_mm >= 2.5
```